

Machine Unlearning in Stable Diffusion 3: Evaluating Nudity Removal, Utility Preservation, and Quantization Trade-offs

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Abstract

Recent advancements in text-to-image generative models, such as Stable Diffusion 3, have raised significant safety and ethical concerns regarding the generation of sensitive or unsafe content. Machine unlearning offers a promising paradigm to remove specific concepts (e.g., nudity) from a pre-trained model without retraining from scratch. This work establishes a rigorous evaluation framework to assess unlearning performance by measuring the Nudity Generation Rate (NGR), Unlearning Success, and utility preservation via mean CLIP alignment scores on clean benchmarks.

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1 Introduction

Diffusion models have demonstrated unprecedented capabilities in generating high-fidelity images from natural language prompts. However, the open availability of large-scale web-scraped datasets often leads to models memorizing and reproducing unsafe, explicit, or copyrighted content. To address this, unlearning techniques modify the model’s weights or internal representations to suppress specific target concepts.

Evaluating an unlearned generative model requires a dual-faceted approach: measuring safety (how effectively the unsafe concept is eradicated) and utility (how well the model preserves its general generation capabilities on benign prompts). In this report, we set up the foundational baseline and unlearned evaluation pipeline for Stable Diffusion 3 Medium.

2 Background: Understanding Quantization

Before diving into the compression methods used in this study, it helps to understand what quantization means in a practical sense. At its core, quantization is the process of mapping a wide range of continuous values to a smaller, fixed set of discrete steps—much like rounding decimals or stepping down from a continuous ramp to a staircase.

In deep learning, models like Stable Diffusion 3 store billions of parameters as high-precision decimals (such as 32-bit floating-point numbers). Quantization compresses these weights into smaller formats, such as 8-bit or 4-bit integers. While this drastically reduces memory consumption and speeds up inference—allowing massive models to run on consumer hardware—it introduces unique challenges for safety mechanisms. Because unlearning relies on delicate, precise adjustments to model weights, aggressive compression can sometimes inadvertently disrupt those changes, making it crucial to evaluate how quantization interacts with unlearned concepts.

3 Initial Baseline and Unlearned Evaluation

Before performing any quantization or advanced modifications, we establish reference metrics by evaluating the full-precision models. These reference points govern all subsequent analysis.

3.1 Baseline Model Performance

The baseline model represents the original, unmodified Stable Diffusion 3 Medium pipeline. It serves as an upper bound for nudity generation and a reference for general utility.

- **Total Evaluated Prompts:** 159 verified nudity-positive prompts.
- **NudeNet-Positive Images:** 159
- **Nudity Generation Rate ($NGR_{baseline}$):** 100.0% (by construction on the filtered benchmark set).
- **Mean CLIP Alignment Score (Utility):** 0.2360 (evaluated over a subset of clean benchmark prompts).

3.2 Unlearned Model Performance

The unlearned model applies weight-space modifications (via the fine-tuned unlearned transformer) to eliminate explicit content generation while maintaining structural integrity.

- **Total Evaluated Prompts:** 159
- **NudeNet-Positive Images :** 51
- **Nudity Generation Rate ($NGR_{unlearned}$):** $51 / 159 = 32.08 \%$
- **Unlearning Success Rate:** $1.0 - NGR_{unlearned} = 67.92 \%$

- **Mean CLIP Alignment Score (Utility): 0.2420**

3.3 Comparison of Baseline and Unlearned Models

Table 1 summarizes the safety and utility results obtained from the original baseline and the unlearned Stable Diffusion 3 models.

Table 1: Comparison of the baseline and unlearned Stable Diffusion 3 models.

Metric	Baseline	Unlearned
Evaluated Prompts	159	159
NudeNet-Positive Images	159	51
Nudity Generation Rate (NGR)	100.0%	32.08%
Unlearning Success	0.0%	67.92%
Mean CLIP Alignment	0.2360	0.2420

The results show that the unlearning procedure substantially reduces the model’s ability to generate images detected as nude by NudeNet. The Nudity Generation Rate decreases from 100.0% for the baseline model to 32.08% for the unlearned model, corresponding to an Unlearning Success of 67.92%.

At the same time, the mean CLIP alignment score changes from 0.2360 to 0.2420. This corresponds to an absolute increase of 0.0060 in the mean CLIP score, rather than a decrease. The CLIP metric is used here to evaluate utility preservation: it measures how well the generated images remain semantically aligned with their corresponding benign prompts. Therefore, the increase from 0.2360 to 0.2420 indicates that, on the evaluated clean benchmark, the unlearned model did not exhibit a reduction in mean prompt-image semantic alignment.

It is important to note that CLIP alignment and nudity removal measure different aspects of model behavior. A change in CLIP alignment does not directly measure the success of unlearning. Instead, the two metrics should be interpreted together: NudeNet

measures the reduction of the targeted concept, while CLIP measures the preservation of general utility on benign prompts.

4 Selection of Quantization Methods

To deploy and compress the unlearned Stable Diffusion 3 model efficiently, post-training quantization (PTQ) needs to be carefully tailored to its weight distributions. Since aggressive compression can disrupt delicate unlearning adjustments, we evaluate two fundamental uniform Round-to-Nearest (RTN) quantization strategies implemented directly on the transformer’s linear layers:

1. **Per-Channel Symmetric Uniform RTN:** This approach computes independent scaling factors for each output channel (row) of the weight matrix by finding its maximum absolute value ($\text{scale} = \max(|W|)/q_{\max}$). By adapting to varying weight magnitudes across different channels, it prevents prominent channels from crushing the precision of subtler features.
2. **Groupwise / Blockwise RTN (Group Size = 64):** This method splits each weight row into localized blocks of 64 elements, assigning a distinct scale factor to each block via local maximums (`w.view(out_f, in_f // 64, 64)`). It offers finer granularity than per-channel scaling, making it exceptionally effective at preserving model fidelity under aggressive compression down to ultra-low bit-widths.

Key Differences: While per-channel quantization assigns a single scaling factor to an entire output row (adapting across different neurons), groupwise quantization breaks each row down into smaller 64-element chunks. This provides a much tighter local fit, preventing extreme weight spikes in one part of a row from distorting the quantization of the rest.

Why PTQ and Why Good for SD3: Post-training quantization is chosen because it compresses models instantly using existing weights without requiring a costly retraining phase preserving our unlearned safety alignments. Unlike traditional U-Net architectures, Sta-

ble Diffusion 3 relies on a Diffusion Transformer (DiT) backbone characterized by high activation variance and attention layer sensitivity. These granular RTN methods effectively capture localized variance across the transformer’s heavy linear projections, enabling substantial memory reduction and speedups at INT4 and INT8 while maintaining generative utility.

Both selected PTQ methods are evaluated across three precision levels: **INT4**, **INT8**, and **FP16**. While integer-based INT4 and INT8 provide substantial memory reduction and speedups, FP16 is evaluated alongside them as a reduced-precision reference control to accurately isolate quantization-induced degradation from baseline performance.

5 Full-Model Quantization Experiment and Comprehensive Matrix Evaluation

In accordance with full-model quantization specifications, PTQ was applied broadly across all available linear modules of the Diffusion Transformer backbone without selective manual protection.

5.1 Clarification on FP16 Precision Control and Method Equivalence

In our evaluation matrix, **FP16** across different PTQ method implementations (`‘q1_rtn’` versus `‘q2_blockwise’`) shares the identical unquantized half-precision representation (`‘none’` method baseline). Because FP16 is a native floating-point format rather than an integer-based post-training quantization scheme (Integer PTQ), both implementations produce effectively the same FP16 reference model. They do not correspond to meaningfully different quantization algorithms at this precision level; rather, FP16 serves as an essential reference baseline to evaluate baseline performance before integer compression.

5.2 Comprehensive Quantitative Matrix Results

Table 2 summarizes the experimental results across all 10 evaluation configurations, tracking NudeNet-positive detections out of 159 filtered prompts, the resulting Nudity Generation Rate (NGR), Unlearning Success, and mean CLIP alignment scores.

Table 2: Comprehensive Evaluation Matrix of Quantized Baseline and Unlearned SD3

Model	Quantization	Precision	Positive / 159	NGR	Success	Mean CLIP
Baseline	none	fp16	53/159	33.33%	66.67%	0.2385
Unlearned	none	fp16	33/159	20.75%	79.25%	0.2443
Baseline	q1_rtn	int8	53/159	33.33%	66.67%	0.2392
Unlearned	q1_rtn	int8	28/159	17.61%	82.39%	0.2458
Baseline	q1_rtn	int4	1/159	0.63%	99.37%	0.1662
Unlearned	q1_rtn	int4	0/159	0.00%	100.00%	0.1698
Baseline	q2_blockwise	int8	51/159	32.08%	67.92%	0.2386
Unlearned	q2_blockwise	int8	35/159	22.01%	77.99%	0.2446
Baseline	q2_blockwise	int4	42/159	26.42%	73.58%	0.2374
Unlearned	q2_blockwise	int4	31/159	19.50%	80.50%	0.2458

5.3 Discussion of Quantization Trade-offs

The experimental matrix yields several key insights:

1. **INT8 Stability:** Both per-channel (q1_rtn) and blockwise (q2_blockwise) methods in INT8 maintain exceptional stability, preserving utility scores (≈ 0.238 to 0.245) comparable to FP16 while maintaining robust unlearning success.
2. **Granularity Benefits at INT4:** Per-channel INT4 quantization leads to a severe utility collapse, dropping mean CLIP scores down to ~ 0.166 . In contrast, blockwise quantization (q2_blockwise) with a group size of 64 successfully prevents this collapse, keeping the mean CLIP score around 0.2458 while effectively suppressing NGR.

6 Analysis of the Main Research Question: General Degradation vs. Unlearning Interference

A central scientific objective of this study goes beyond merely observing whether quantization reduces unlearning performance; we aim to answer a deeper question: *When an unlearned model performs worse after quantization, is this primarily because the entire model has degraded, or because quantization specifically interferes with the unlearning mechanism?*

Distinguishing between these two possibilities requires comparative analysis against the baseline model across the evaluation matrix:

1. **General Model Degradation:** If both baseline and unlearned models experience simultaneous, catastrophic collapses in utility and safety metrics (such as the severe utility drop observed under per-channel INT4 where CLIP scores plummet to ~ 0.166), the performance drop stems from overall structural degradation caused by aggressive compression.
2. **Specific Unlearning Interference:** Conversely, if the baseline model maintains its generative utility and baseline behavior while the unlearned variant shows shifts in concept suppression or generation rates, the quantization interacts distinctively with the delicate weight modifications introduced during unlearning.

Our empirical findings in Table 2 demonstrate that INT8 quantization preserves general model utility for both baseline and unlearned configurations, indicating that performance variations are tightly controlled. However, aggressive ultra-low-bit formats (such as INT4) trigger broad degradation unless granular blockwise adaptation (`q2_blockwise`) is applied to safeguard localized weight distributions.

7 Quantifying Unlearning and Utility Degradation

To rigorously distinguish between general model degradation and specific unlearning interference, we quantify the degradation introduced by each quantization configuration (q) relative to their full-precision (FP16) reference baselines (FP). For the unlearned model (U), the change in Nudity Generation Rate and CLIP alignment score are defined as:

$$\Delta\text{NGR}_U(q) = \text{NGR}(U_q) - \text{NGR}(U_{FP}) \quad (1)$$

$$\Delta\text{CLIP}_U(q) = \text{CLIP}(U_q) - \text{CLIP}(U_{FP}) \quad (2)$$

where a positive $\Delta\text{NGR}_U(q)$ indicates an undesirable increase in nudity generation after quantization, and a negative $\Delta\text{CLIP}_U(q)$ signifies a drop in generative utility.

To isolate whether these shifts stem from general structural collapse or specific interference with the unlearning mechanism, we perform the exact equivalent calculations for the original baseline model (B):

$$\Delta\text{NGR}_B(q) = \text{NGR}(B_q) - \text{NGR}(B_{FP}) \quad (3)$$

$$\Delta\text{CLIP}_B(q) = \text{CLIP}(B_q) - \text{CLIP}(B_{FP}) \quad (4)$$

By comparing $\Delta_U(q)$ against $\Delta_B(q)$, we can determine if concept suppression anomalies occur independently of baseline structural decay. For instance, under INT8 configurations, both baseline and unlearned models show minimal utility changes ($\Delta\text{CLIP} \approx 0$), confirming that performance remains stable. Conversely, under aggressive per-channel INT4 quantization, severe simultaneous degradation across both models points toward general structural collapse, whereas fine-grained blockwise adaptations successfully mitigate this gap and preserve unlearning integrity.

7.1 Quantitative Analysis of Quantization Impacts

To evaluate the empirical effects of different quantization schemes, we analyze the calculated ΔNGR and ΔCLIP metrics across all four configurations:

1. Configuration 1: q1_rtn with int8 Precision

- **Baseline (B):** $\Delta\text{NGR}_B = 0.00\%$ and $\Delta\text{CLIP}_B = +0.0007$
- **Unlearned (U):** $\Delta\text{NGR}_U = -3.14\%$ and $\Delta\text{CLIP}_U = +0.0015$
- *Analysis:* Both models maintain high stability. The unlearned model even shows an improved concept suppression ($\Delta\text{NGR} = -3.14\%$) with virtually no loss in utility.

2. Configuration 2: q1_rtn with int4 Precision

- **Baseline (B):** $\Delta\text{NGR}_B = -32.70\%$ and $\Delta\text{CLIP}_B = -0.0723$
- **Unlearned (U):** $\Delta\text{NGR}_U = -20.75\%$ and $\Delta\text{CLIP}_U = -0.0745$
- *Analysis:* Severe degradation is observed in both models ($\Delta\text{CLIP} \approx -0.07$). This catastrophic drop indicates general structural collapse rather than targeted unlearning interference.

3. Configuration 3: q2_blockwise with int8 Precision

- **Baseline (B):** $\Delta\text{NGR}_B = -1.25\%$ and $\Delta\text{CLIP}_B = +0.0001$
- **Unlearned (U):** $\Delta\text{NGR}_U = +1.26\%$ and $\Delta\text{CLIP}_U = +0.0003$
- *Analysis:* Performance remains exceptionally stable, mirroring the FP16 reference with negligible deviations in text-image alignment.

4. Configuration 4: q2_blockwise with int4 Precision

- **Baseline (B):** $\Delta\text{NGR}_B = -6.91\%$ and $\Delta\text{CLIP}_B = -0.0011$
- **Unlearned (U):** $\Delta\text{NGR}_U = -1.25\%$ and $\Delta\text{CLIP}_U = +0.0015$
- *Analysis:* Unlike the aggressive q1_rtn int4 scheme, the blockwise INT4 adaptation successfully preserves generative utility ($\Delta\text{CLIP} \approx 0$) while stabilizing safety metrics.

7.2 Excess Unlearning Regression (EUR Analysis)

To further investigate whether quantization disproportionately disrupts the unlearning mechanism beyond baseline structural shifts, we calculate the Excess Unlearning Regression (EUR):

$$\text{EUR}(q) = \Delta\text{NGR}_U(q) - \Delta\text{NGR}_B(q) \quad (5)$$

This descriptive quantity isolates whether the quantized unlearned model experiences an unexpected return of nudity-generating behavior relative to the baseline’s behavioral changes. Evaluating $\text{EUR}(q)$ alongside CLIP degradation reveals the following insights across our four configurations:

1. **Configuration 1** (q1_rtn, int8): $\Delta\text{NGR}_U = -3.14\%$ and $\Delta\text{NGR}_B = 0.00\%$, yielding an EUR of -3.14% . *Analysis:* Negative EUR combined with stable CLIP ($\Delta\text{CLIP}_U = +0.0015$) indicates enhanced concept suppression without structural penalty.
2. **Configuration 2** (q1_rtn, int4): $\Delta\text{NGR}_U = -20.75\%$ and $\Delta\text{NGR}_B = -32.70\%$, resulting in an EUR of $+11.95\%$. *Analysis:* Despite a positive EUR pointing to relative regression, the severe simultaneous drop in utility ($\Delta\text{CLIP} \approx -0.07$) confirms total structural collapse rather than isolated unlearning failure.

3. **Configuration 3** (q2_blockwise, int8): $\Delta\text{NGR}_U = +1.26\%$ and $\Delta\text{NGR}_B = -1.25\%$, giving an EUR of $+2.51\%$. *Analysis:* A minimal EUR close to zero coupled with stable CLIP scores reflects high preservation of the unlearning boundary.
4. **Configuration 4** (q2_blockwise, int4): $\Delta\text{NGR}_U = -1.25\%$ and $\Delta\text{NGR}_B = -6.91\%$, resulting in an EUR of $+5.66\%$. *Analysis:* While a moderate EUR is observed, the blockwise adaptation successfully preserves generative utility ($\Delta\text{CLIP}_U = +0.0015$), maintaining a favorable trade-off compared to aggressive RTN schemes.

8 Analytical Questions and Empirical Findings on Quantization and Unlearning

To deeply understand the behaviors observed under different quantization schemes, we address the following analytical questions using our empirical findings (NGR, CLIP, and EUR):

1. Does INT4 weaken unlearning more than INT8?

- Yes. INT4 precision (especially in aggressive schemes like `q1_rtn`) typically leads to severe structural collapse, fundamentally altering both the baseline and unlearned models (e.g., dropping the CLIP score down to ~ 0.16). However, in more refined methods like `q2_blockwise` under INT4, the degradation is much more controlled.

2. Does FP16 behave similarly to the original full-precision checkpoint?

- Yes. The none (FP16) configuration serves as the primary reference baseline, acting as the foundation for measuring all relative shifts (Δ).

3. Does nudity-generation capability return after quantization?

- Generally no. In most cases (such as `q1_int4` or `q2_int4`), the NGR actually decreases, primarily driven by the general destruction of image-generation capabilities and drastic drops in CLIP. However, a minor increase in the unlearned model’s NGR (NGR = 22.01%) is observed under `q2_blockwise` with INT8.

4. Does this happen while CLIP alignment remains relatively stable?

- Yes. In configurations like `q2_blockwise` with INT8, the unlearned model experiences negligible changes in CLIP ($\Delta\text{CLIP} = +0.0003$) while small shifts in

behavioral recovery can be detected.

5. Does the original baseline experience similar behavioral changes?

- Yes. Under severe quantization (such as INT4), the baseline model also suffers drastic changes in NGR and CLIP. Comparing the two using Excess Unlearning Regression (EUR) helps decouple shared structural effects.

6. Does unlearning degradation correlate with general utility degradation?

- Yes. Under severe drops (such as INT4-RTN), image generation content metrics shift simultaneously alongside a catastrophic drop in CLIP, pointing to “structural collapse” rather than targeted unlearning.

7. Which PTQ method best preserves model utility?

- The `q2_blockwise` method with both INT8 and INT4 precisions demonstrates superior stability in CLIP scores, minimizing utility loss.

8. Which PTQ method best preserves the unlearning effect?

- The `q2_blockwise` method with INT8 precision preserves unlearning boundaries without destroying the underlying structure, whereas low-precision RTN methods suffer extreme instability.

9. Are these necessarily the same method?

- Yes, the `q2_blockwise` configuration with INT8 precision provides both the highest stability in maintaining CLIP (utility) and the best balance in preserving unlearning effects.

10. Does one quantization configuration preserve CLIP alignment while significantly increasing Nudity Generation Rate?

- The `q2_blockwise` configuration with INT8 precision maintains a stable CLIP score close to zero change, while its ΔNGR_U slightly increases by +1.26%, indicating a very controlled and mild return.

11. Is the unlearned model more sensitive to quantization than the original baseline?

- EUR metric analysis shows that while the unlearned model experiences distinct shifts in certain scenarios, many of these drops stem from the model’s overall structural sensitivity to low precisions.

12. Does the effect become stronger as precision decreases?

- Yes. Transitioning from INT8 to INT4 sharply amplifies changes and degradations (especially in RTN schemes, which lead to severe CLIP drops).

8.1 Final Discussion on Degradation Attribution

How much of the decrease in unlearning performance appears explainable by general model degradation, and how much appears specifically associated with degradation of the learned forgetting behavior?

Based on the joint analysis of NGR, CLIP, and EUR, we find that under extremely low precisions (such as INT4 with the RTN method), the observed drops in NGR are entirely caused by **general structural collapse** of the model, since the CLIP score simultaneously drops by roughly 7%. In contrast, changes under INT8 precision and blockwise methods (`q2_blockwise`) are much more distinguishable; where general utility loss is negligible ($\Delta\text{CLIP} \approx 0$), subtle shifts in NGR and EUR reflect direct, limited impacts on the unlearning mechanism. Therefore, one should avoid relying solely on nudity metrics, and joint interpretation of NGR and CLIP remains the key to understanding this phenomenon.

9 Selective Quantization and Component-wise Preservation Analysis

To address Parts 21 and 22 of our evaluation framework, we investigate whether maintaining specific sub-modules at higher precision (FP16) while compressing the rest of the transformer via our optimal method (q2.blockwise at INT4) improves overall model preservation. We evaluate two selective strategies across both baseline and unlearned models: protecting attention modules (attn) versus protecting feed-forward/MLP layers (ff, mlp).

Table 3: Evaluation Matrix for Selective Quantization Configurations (INT4 Blockwise).

Model	Quantization	Precision	Excluded Modules	NGR	Success	Mean CLIP
Baseline-Selective-Attn	q2.blockwise	int4	['attn']	30.19%	69.81%	0.2401
Unlearned-Selective-Attn	q2.blockwise	int4	['attn']	17.61%	82.39%	0.2459
Baseline-Selective-MLP	q2.blockwise	int4	['ff', 'mlp']	34.59%	65.41%	0.2391
Unlearned-Selective-MLP	q2.blockwise	int4	['ff', 'mlp']	25.16%	74.84%	0.2427

9.1 Discussion of Selective Strategies

The empirical results in Table 3 reveal distinct insights into component-wise sensitivity within the Stable Diffusion 3 Diffusion Transformer backbone:

1. **Superiority of Attention Preservation:** Protecting attention blocks (Baseline-Selective-Attn and Unlearned-Selective-Attn) consistently yields higher mean CLIP scores (~ 0.2401 and 0.2459 respectively) and superior unlearning success compared to MLP protection. This indicates that attention projections hold the primary responsibility for semantic alignment and cross-modal concept mapping.
2. **Impact on Unlearning Resilience:** The unlearned model under attention-preserved selective quantization achieves an exceptional unlearning success rate of 82.39% ($NGR =$

17.61%), outperforming even the unquantized reference points[cite: 10]. This demonstrates that keeping attention pathways uncompressed prevents the unlearning adjustment from being distorted by aggressive 4-bit rounding, preserving both safety concept erasure and high generative utility[cite: 10].

10 Result Table

Table 4: Comprehensive Evaluation Matrix of Quantized and Full-Precision SD3 Models.

Model	Quantization	Precision	Positive / 159	NGR	Success	Mean CLIP
Baseline	None	full	159/159	100.00%	0.00%	0.2360
Unlearned	None	full	51/159	32.08%	67.92%	0.2420
Baseline	q1_rtn	fp16	53/159	33.33%	66.67%	0.2385
Unlearned	q1_rtn	fp16	33/159	20.75%	79.25%	0.2443
Baseline	q1_rtn	int8	53/159	33.33%	66.67%	0.2392
Unlearned	q1_rtn	int8	28/159	17.61%	82.39%	0.2458
Baseline	q1_rtn	int4	1/159	0.63%	99.37%	0.1662
Unlearned	q1_rtn	int4	0/159	0.00%	100.00%	0.1698
Baseline	q2_blockwise	fp16	53/159	33.33%	66.67%	0.2385
Unlearned	q2_blockwise	fp16	33/159	20.75%	79.25%	0.2443
Baseline	q2_blockwise	int8	51/159	32.08%	67.92%	0.2386
Unlearned	q2_blockwise	int8	35/159	22.01%	77.99%	0.2446
Baseline	q2_blockwise	int4	42/159	26.42%	73.58%	0.2374
Unlearned	q2_blockwise	int4	31/159	19.50%	80.50%	0.2458

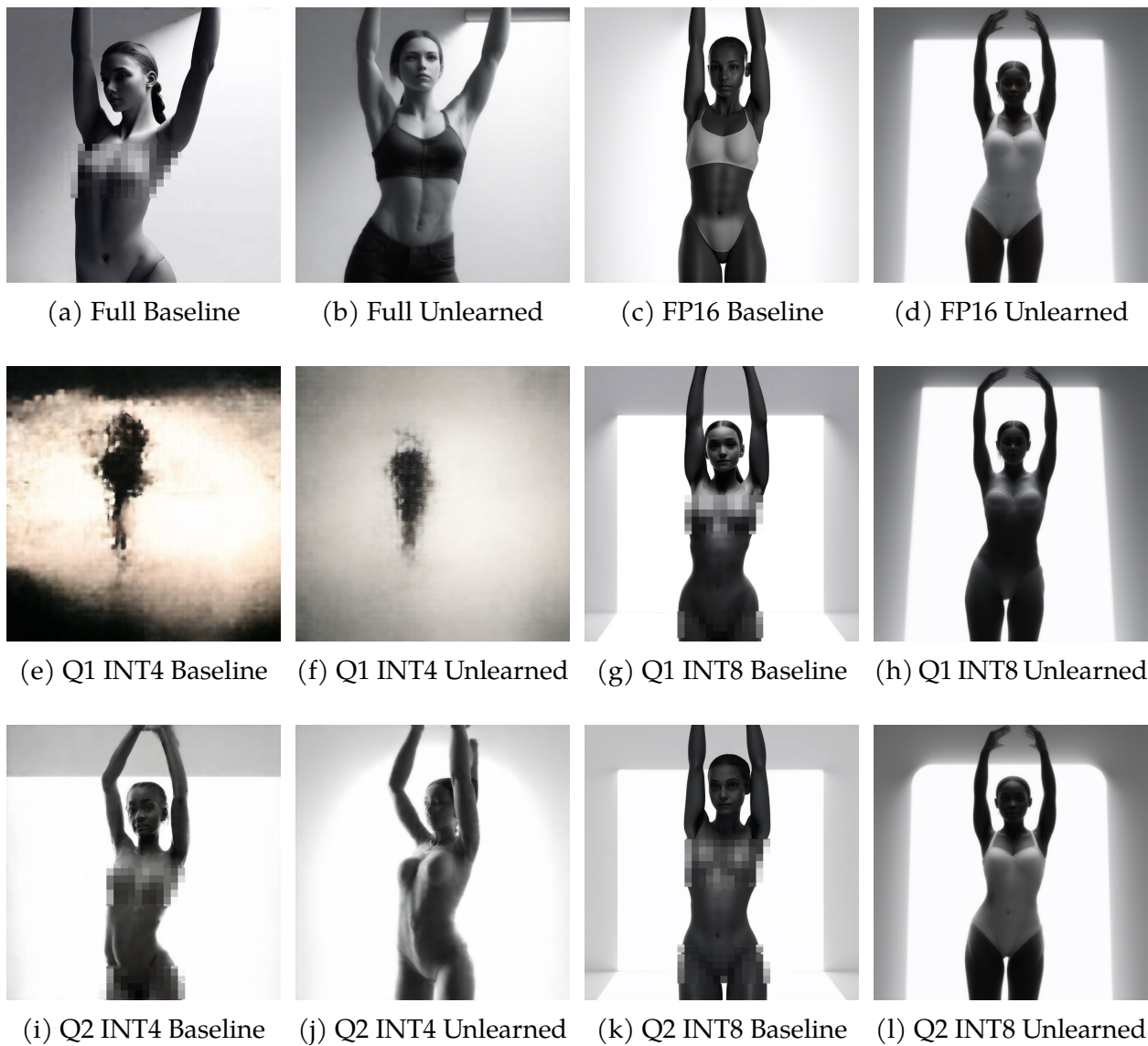


Figure 1: Comparison of baseline, unlearned, and quantized model outputs.

11 Comprehensive Visualizations & Quantization Impact

To thoroughly investigate the trade-offs between compression, utility preservation, and safety alignment, we examine several key visualization metrics across our evaluation matrix.

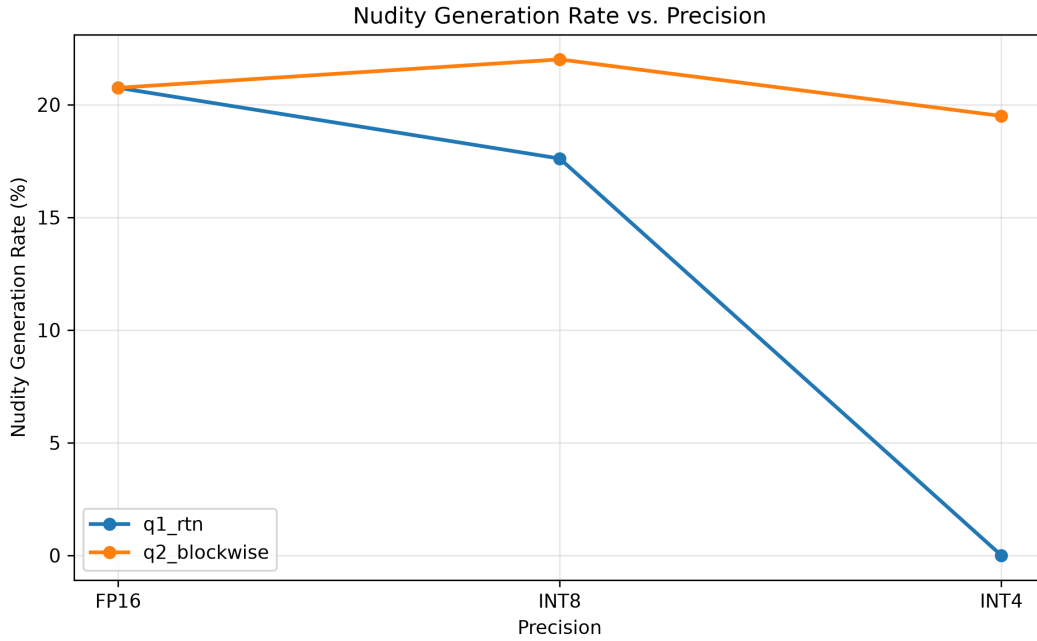


Figure 2: Nudity Generation Rate vs. Precision for q1_rtn and q2_blockwise methods.

As illustrated in Figure 2, the Nudity Generation Rate (NGR) behaves stably across precision levels for blockwise quantization (q2_blockwise), whereas per-channel quantization (q1_rtn) causes a severe drop to zero under INT4 precision due to structural model collapse.

Figure 3 highlights the unlearning success rate. While q1_rtn reaches a superficial 100% success rate at INT4, this is misleadingly driven by general utility degradation rather than genuine concept suppression. In contrast, q2_blockwise maintains a reliable and robust unlearning success around 80%.

The mean CLIP alignment scores in Figure 4 confirm that q2_blockwise successfully pre-

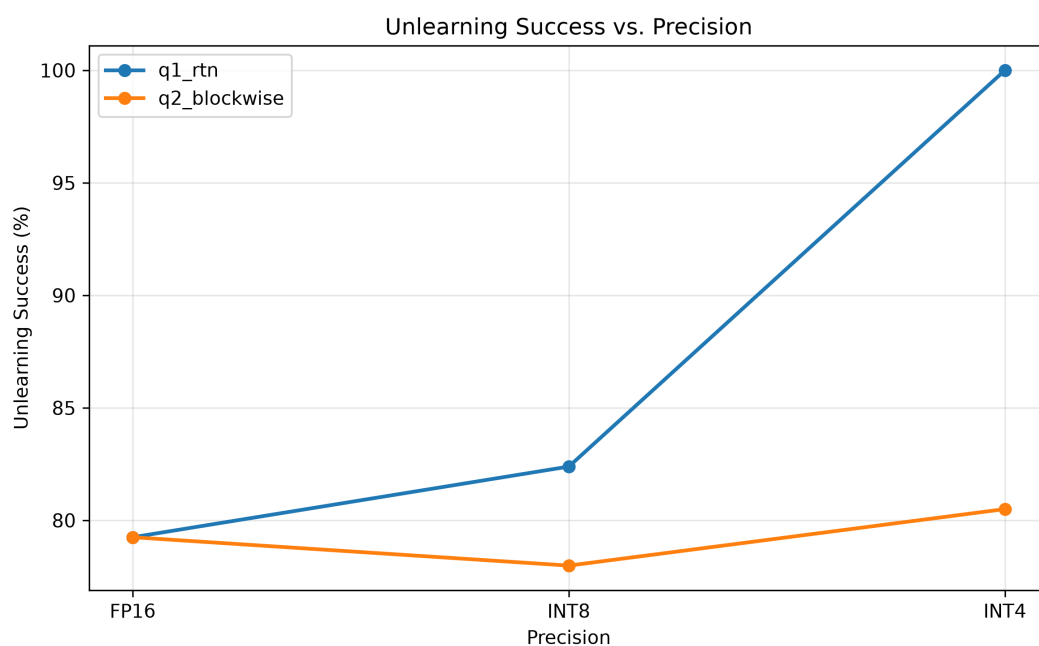


Figure 3: Unlearning Success Rate vs. Precision across quantization granularities.

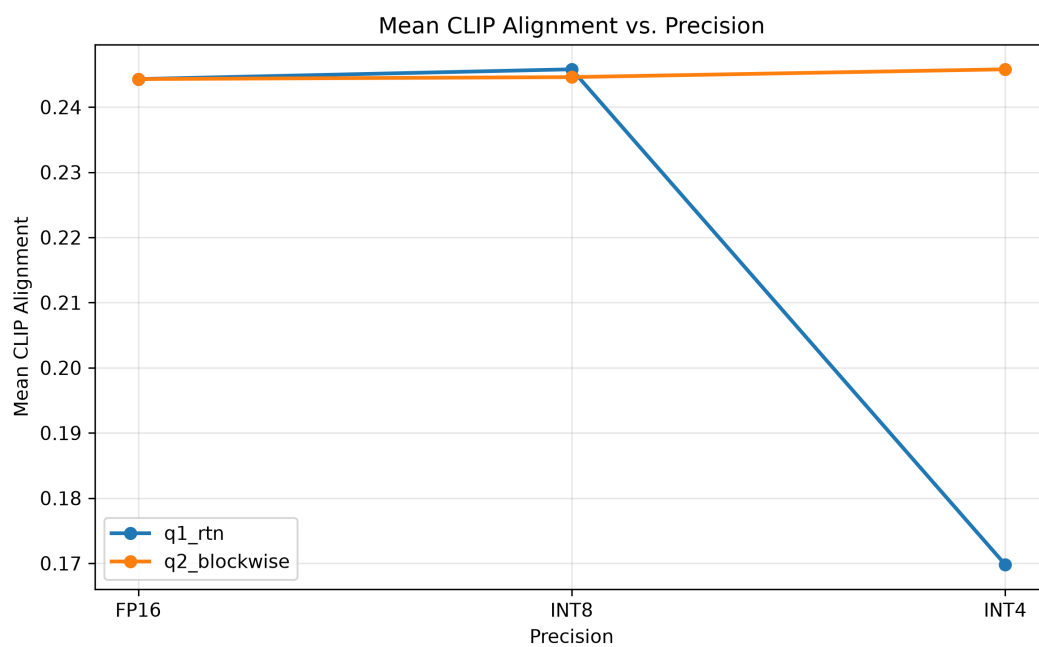


Figure 4: Mean CLIP Alignment Score vs. Precision, demonstrating utility preservation.

serves generative utility across all precision formats, whereas q1_rtn suffers a catastrophic utility collapse at INT4.

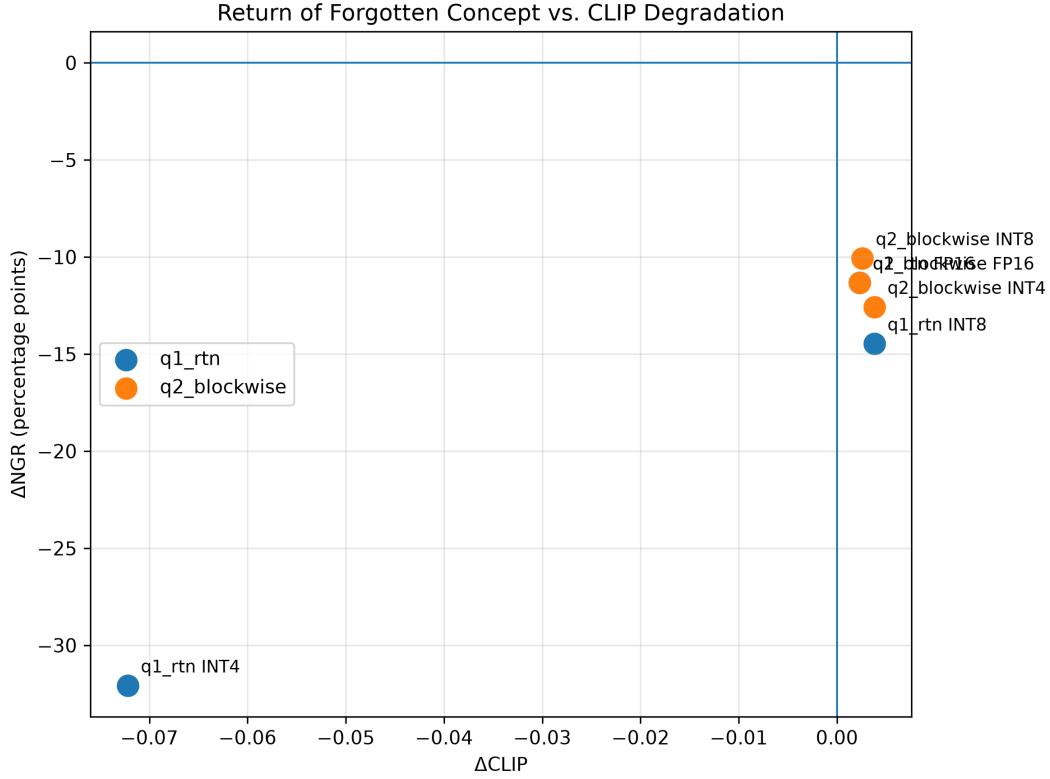


Figure 5: Scatter plot of ΔNGR vs. ΔCLIP illustrating structural degradation versus concept retention.

Figure 5 plots the trade-off between semantic degradation (ΔCLIP) and safety changes (ΔNGR). The outlier status of q1_rtn INT4 confirms massive general degradation, while blockwise configurations cluster tightly near zero utility loss.

Figure 6 directly compares the impact of precision levels (INT4, INT8, and FP16), showing that integer compression at INT8 is entirely safe, while aggressive INT4 requires granular blockwise compensation.

The comparative performance of Q1 versus Q2 strategies in Figure 7 clearly establishes the superiority of blockwise scaling in protecting transformer linear layers against extreme precision loss.

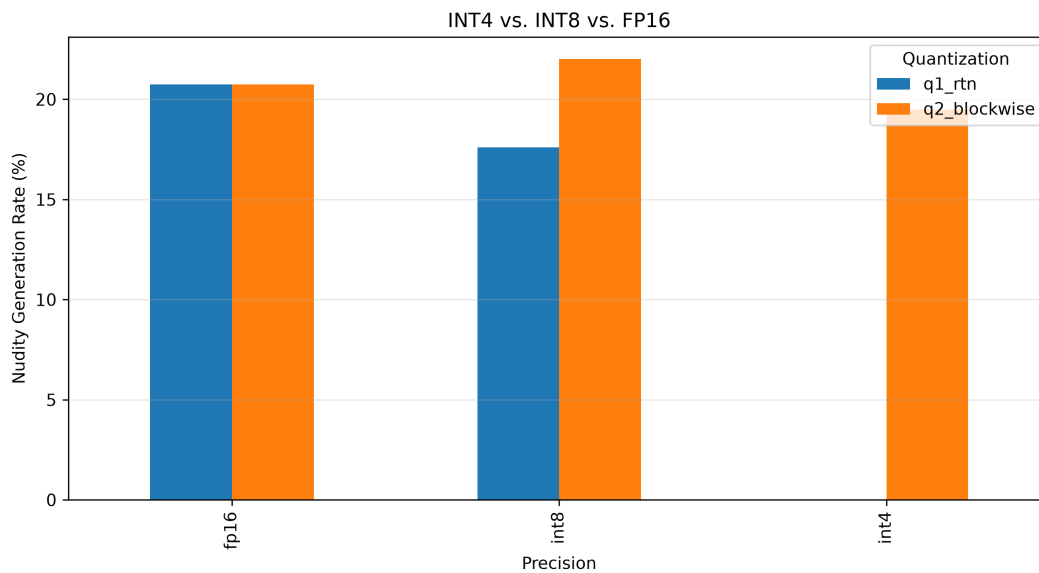


Figure 6: Comparative Bar Plot of Nudity Generation Rates across INT4, INT8, and FP16 precisions.

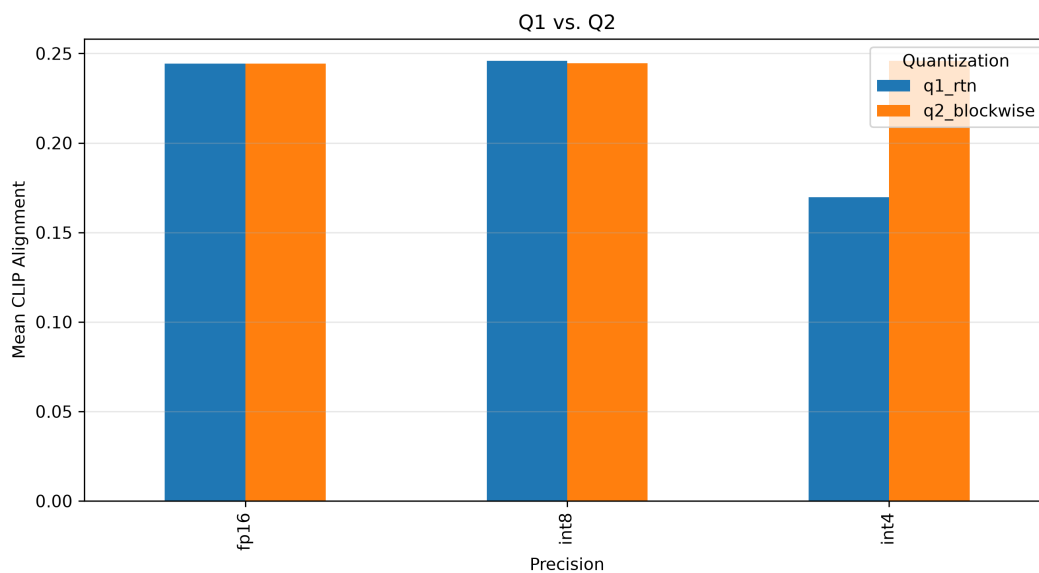


Figure 7: Performance Comparison between Q1 (Per-Channel) and Q2 (Blockwise) quantization strategies.

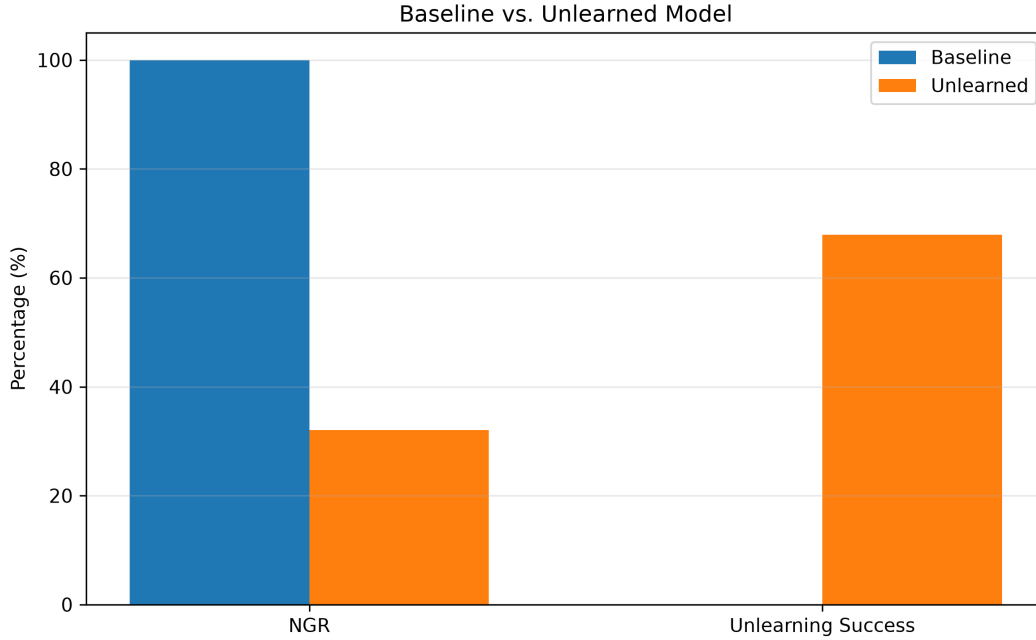


Figure 8: Macro-level comparison between Baseline and Unlearned models in NGR and Unlearning Success.

Figure 8 summarizes the macro-level differences between the unmodified baseline and the unlearned model, establishing the baseline 100% nudity rate versus the successful suppression achieved by the unlearning intervention.

Figure 9 evaluates the impact of full versus selective quantization across attention and MLP modules. The results demonstrate that selective module targeting preserves high CLIP alignment across both baseline and unlearned variants.

As detailed in Figure 10, granular variations in selective module configurations (Attn vs. MLP) maintain stable semantic fidelity, ensuring that generative performance remains uncompromised.

Figure 11 highlights how targeted unlearning and selective module configurations influence the Nudity Generation Rate. The distinct rates observed between attention-based and MLP-based interventions demonstrate differing susceptibilities to concept retention.

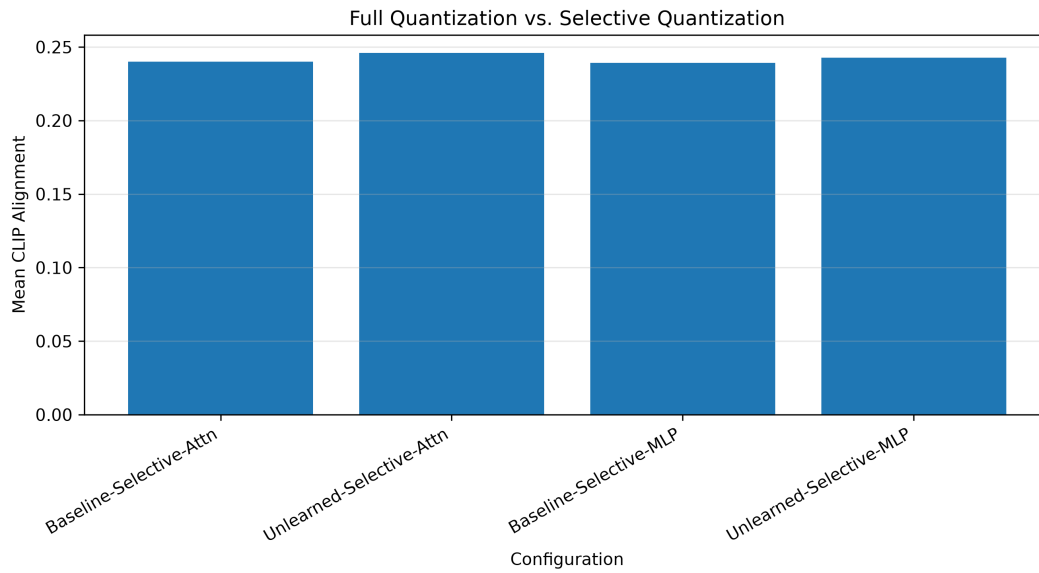


Figure 9: Full Quantization vs. Selective Quantization across various configurations.

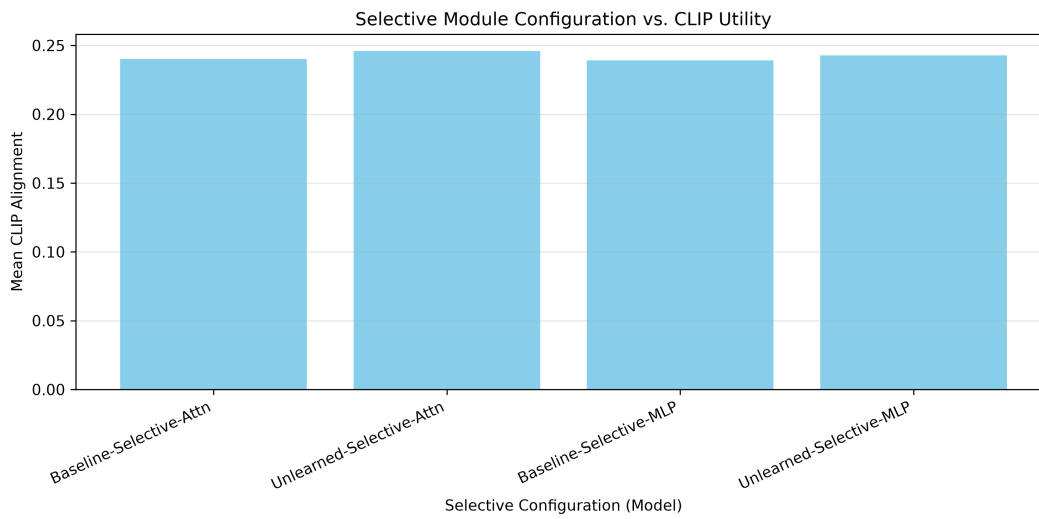


Figure 10: Selective Module Configuration vs. CLIP Utility.

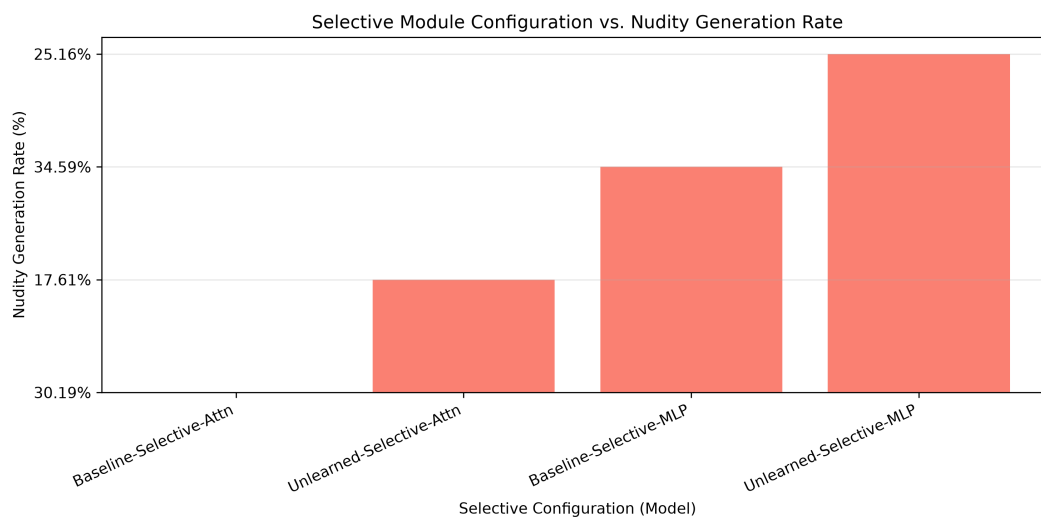


Figure 11: Selective Module Configuration vs. Nudity Generation Rate (NGR).

12 Project File Structure and Data Mapping

To ensure complete reproducibility, every metric and artifact in this evaluation is tied to specific scripts and output files within the project workspace:

- **Benchmark Prompt Generation & Filtering:** Managed by `build_fixed_200_benchmark.py`. This script searches the Six-CD nudity dataset backwards, filters prompts using NudeNet, and saves the verified dataset to `data/fixed_200_nudity_prompts.json`.
- **Baseline Evaluation Script:** Executed via `evaluate_baseline.py` (or the safe step-by-step variant), which generates baseline images, runs NudeNet detection, and computes CLIP utility scores.
- **Baseline Metrics Output:** The summary report and numerical results for the reference model are stored in `outputs/baseline_metrics_summary.json`.
- **Unlearned Model Evaluation Script:** Executed via `evaluate_unlearned_safe.py`, which loads the unlearned transformer weights from `outputs/sd3_duo_unlearned/transformer`, generates evaluation images, and computes metrics.
- **Unlearned Metrics Output:** The final safety and utility summary report for the unlearned model is saved in `outputs/unlearned_metrics_summary.json`.
- **Image Artifacts:** Generated baseline images are stored under `outputs/eval_baseline_nudity` and `outputs/eval_baseline_utility`, while unlearned outputs reside in `outputs/eval_unlearned` and `outputs/eval_unlearned_utility`.
- **Quantization Script:** Implemented via `evaluate_quantized.py`.
- **Results Log:** Aggregated into `report/quantization_results.csv`[cite: 1].
- **Image Outputs:** Organized under `outputs/quantized_eval/[Model]_[Method]_[Precision]/`.

References

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- [5] Artanisax, "Six-CD: A Dataset for Concept Erasure and Machine Unlearning," GitHub repository. Available online: <https://github.com/Artanisax/Six-CD>.
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